

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B1. FLUIDS

TRINITY TERM 2022

Friday, 10th June, 2:30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

The Navier–Stokes equation for viscous, incompressible, fluid flow under gravity is

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \frac{1}{\rho} \nabla p + g \mathbf{k} = \nu \nabla^2 \mathbf{u} ,$$

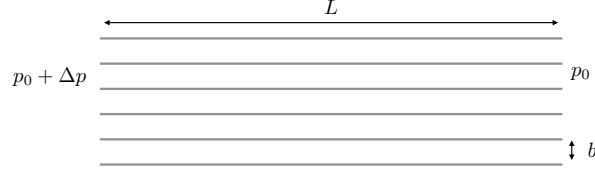
where $\mathbf{u}$ is the fluid velocity, ρ the density, p the pressure, g the acceleration due to gravity, $\mathbf{k}$ the vertical unit vector and ν the kinematic viscosity.

The *Jacobian* $\mathbf{J}$ of a dynamical system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ is defined as $J_{ij} = \frac{\partial f_i}{\partial x_j}$, where f_i and x_j are the i^{th} and j^{th} components of $\mathbf{f}(\mathbf{x})$ and $\mathbf{x}$ respectively. The trace τ of the Jacobian is equal to the sum of its eigenvalues, while the determinant Δ is equal to their product.

For problems in atmospheric physics, unless otherwise stated, x and y refer to the horizontal eastward and northward directions and z the upward direction with corresponding velocities u , v and w .

$$1000 \text{ mb} = 1000 \text{ hPa} = 10^5 \text{ Pa}.$$

1. (a) Consider a porous manifold comprising of n parallel thin rectangular sheets of rigid solid with negligible thickness, length L and large width w , with adjacent sheets separated by distance $b \ll w, L$, as illustrated in the figure. The manifold is filled with incompressible fluid of constant density ρ and dynamic viscosity η . A pressure difference Δp is applied across the length L of the plates, with the pressure gradient constant.



Starting from the Navier-Stokes equations, determine all components of the velocity between an adjacent pair of plates, clearly stating any assumptions made. Hence show that the volume flux per unit cross sectional area satisfies the so-called Darcy's law,

$$q = \frac{\Pi_0}{\eta} \frac{\Delta p}{L},$$

where Π_0 is a constant whose value your answer should specify.

[9]

Explain whether the fluid inertia provides an important contribution to the dominant force balance for this flow.

[2]

(b) A different porous material consists of fluid of constant density in a solid matrix which permits flow in all directions. The generalisation of the above results for Darcy's law yields a directed volume flux per unit cross sectional area, $\mathbf{q}$, satisfying

$$\frac{\eta}{\Pi_1} \mathbf{q} = -\nabla p - \rho g \hat{\mathbf{z}},$$

for constant permeability Π_1 , pressure p , gravitational acceleration g and unit vector $\hat{\mathbf{z}}$. By considering mass conservation in an arbitrary volume, V , large enough to contain many liquid-solid boundaries, show that $\nabla \cdot (\rho \mathbf{q}) = 0$.

[3]

(c) A method for the geological sequestration of carbon dioxide pumps incompressible fluid from a spherical source of radius a held at pressure $p_0 + \Delta p$ embedded at the centre of a large aquifer consisting of porous rock with constant permeability Π_1 filled with fluid. The pressure far from the injection site is p_0 . For the case where all of the fluids have the same constant density and viscosity and the injection pressure is large enough for the effects of gravity to be neglected, determine the radial component of $\mathbf{q}$ as a function of distance r from the source, a , Δp , Π_1 and η .

Consider two different aquifers. Due to differences in the background pressure, the viscosity η_A in aquifer A is twice as large as the viscosity η_B in aquifer B. The pore lengthscale d characterises the characteristic mean distance between solid grains, and is twice as large in aquifer A as it is in aquifer B. Discuss which aquifer would be preferable if one wants to reduce the energy used in injecting the fluid into the porous rock, clearly stating any assumptions you make.

[8]

(d) Discuss how the result to part (c) would change if the gravitational acceleration were non-negligible, and the constant background pressure p_0 were replaced with a vertically varying hydrostatic pressure $p_0(z)$.

[3]

2. (a) With reference to the example of flow of water in a river, explain the difference between the Eulerian and Lagrangian accelerations of a fluid. [3]

(b) Consider a stagnation point flow above a rigid planar boundary at $y = 0$ given by the stream function $\psi = -Axy$ where $A > 0$ is constant, and the gravitational acceleration $\mathbf{g} = -g\hat{\mathbf{y}}$ is constant. Show that the velocity satisfies the boundary conditions required at $y = 0$ for an inviscid flow. By considering the inviscid limit of the Navier-Stokes equations, show that the pressure p above the plate satisfies

$$p = p_\infty(x, y) = p_0 - \frac{1}{2}A^2(x^2 + y^2) - \rho gy,$$

where p_0 is a constant reference pressure. [4]

(c) If the flow instead has a large but finite Reynolds number, explain why a viscous boundary layer must form above the boundary $y = 0$.

Starting from the Navier-Stokes equations for a laminar and incompressible flow and describing any scaling arguments used, derive the boundary layer equations

$$\mathbf{v} \cdot \nabla v_x = \nu \frac{\partial^2 v_x}{\partial y^2} - \frac{1}{\rho} \frac{\partial p_\infty}{\partial x}, \quad \frac{\partial p}{\partial y} = \frac{\partial p_\infty}{\partial y}, \quad \nabla \cdot \mathbf{v} = 0,$$

for velocity components $\mathbf{v} = (v_x, v_y)$, pressure p , density ρ and a constant reference pressure p_0 , where ν is the viscosity. [7]

(d) Look for solutions of the form $\psi = -Ax\delta f(\eta)$, where $\eta = y/\delta$, with δ independent of x and y . Show that with a suitable choice of δ the function $f(\eta)$ satisfies the differential equation

$$\frac{d^3 f}{d\eta^3} + f \frac{d^2 f}{d\eta^2} - \left(\frac{df}{d\eta} \right)^2 = -1$$

and determine the boundary conditions on $f(\eta)$. [6]

(e) Without solving this equation for f , determine an expression for the approximate boundary layer thickness in terms of A , x , y , ν , ρ and p_0 . Discuss what physical effects control the boundary layer thickness.

Estimate the horizontal length-scale l where the boundary layer may become turbulent for $x > l$. [5]

3. Consider a cylindrical tank of radius $0 < r < a$ and depth $-h < z < h$, closed on all sides by impermeable walls. The tank is filled with 2 layers of inviscid, incompressible fluid initially separated by a membrane, with freshwater with density ρ_1 for $z < 0$ and saltwater with larger density ρ_2 for $z > 0$. There is no surface tension between the two layers, and the gravitational acceleration is constant $\mathbf{g} = -g\hat{\mathbf{z}}$. We initially neglect the diffusion of salt. At time $t = 0$ the membrane is burst in a way that causes a purely axisymmetric deformation of the interface $z = \eta(r, t)$ between the fresh and saline layers, with negligible initial velocity.

(a) Explain why the flow in each layer can be described using a velocity potential ϕ_i with $i = 1, 2$ for all times $t \geq 0$, and show that $\nabla^2 \phi_i = 0$ in each region.

If $\eta \ll a, h$, determine an appropriate set of linearised boundary conditions on each ϕ_i , and a boundary condition for continuity of the normal stress involving $\eta(r, t)$. [9]

(b) The axially symmetric solutions of $\nabla^2 f = 0$ that are bounded at the origin take the form

$$f(r, z) = J_0(\alpha r) [A \exp(\alpha z) + B \exp(-\alpha z)],$$

where J_0 is the zeroth-order Bessel function. Look for solutions where the interface displacement takes the form $\eta = \eta_0 \exp(\sigma_n t) J_0(\alpha_n r)$. Determine a condition on the allowed set of values for α_n , and find the corresponding values of σ_n .

Discuss how the stability of the system varies as a function of α_n . [9]

(c) The corresponding problem for two-dimensional flow in a tank of width $0 < x < b$ and depth $-h < z < h$ has modes with growth rates σ_n satisfying

$$\sigma_n^2 = \left(\frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} \right) g k_n \tanh k_n h, \quad k_n = n\pi/b, \quad n = 1, 2, 3, \dots$$

To understand the instability mechanism, we consider the vorticity equation for two-dimensional flow given by

$$\frac{\partial \omega}{\partial t} + \mathbf{v} \cdot \nabla \omega = \nabla \times \left(\frac{\mathbf{F}}{\rho} \right),$$

where the density $\rho(\mathbf{x}, t)$ can vary with space and time, ω is the vorticity, and $\mathbf{F}$ is the sum of forces arising from hydrodynamical stresses and body forces. Show that

$$\frac{\partial \omega}{\partial t} + \mathbf{v} \cdot \nabla \omega = \frac{\nabla \rho \times \nabla p}{\rho^2}.$$

Accounting for weak molecular diffusion of salt, the initially discontinuous jump in the background density profile is replaced by a continuous but rapid transition of the density about $z = 0$ over a small lengthscale $\delta \ll b, h$, where there can also be rapidly varying vorticity. The sharp interface problem from parts (a) and (b) corresponds to letting $\delta \rightarrow 0$ to recover a vortex sheet at $z = \eta$. Using the vorticity equation and appropriate sketches, explain how the vorticity is amplified within the region of rapidly varying vorticity. Hence discuss the physical mechanisms underlying the variation of σ_n with n . [7]

4. (a) Explain the conditions required for a low Reynolds number flow (i.e. $\text{Re} \ll 1$) and give two examples of applications involving low Reynolds number flows.

Consider flows with $\text{Re} \ll 1$, constant density ρ and constant dynamic viscosity η . Starting from the Navier-Stokes equations, show that the vorticity ω and pressure p satisfy $\nabla^2 \omega \approx 0$ and $\nabla^2 p \approx 0$. [7]

(b) Consider a spherical air bubble of radius a rising at constant speed U through water with $\text{Re} \ll 1$. The air bubble has negligible viscosity, the water has non-zero dynamic viscosity η and the effects of surface tension and hydrostatic pressure differences can be neglected in the water.

Consider the reference frame where the far field fluid is at rest, and the bubble moves. At a fixed instant of time t we use spherical co-ordinates (r, θ, ϕ) with the origin at the centre of the bubble and $\theta = 0$ points in the direction of bubble rise. The instantaneous velocity components can be written as

$$v_r = \frac{1}{r^2 \sin \theta} \frac{\partial \Psi}{\partial \theta}, \quad v_\theta = -\frac{1}{r \sin \theta} \frac{\partial \Psi}{\partial r}$$

for some function $\Psi(r, \theta)$. For axisymmetric flows in spherical co-ordinates, the components of the strain rate tensor are

$$e_{rr} = \frac{\partial v_r}{\partial r}, \quad e_{r\theta} = \frac{r}{2} \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{2r} \frac{\partial v_r}{\partial \theta}, \quad e_{\theta\theta} = \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}.$$

Show that the above velocity is incompressible. Determine the boundary conditions on v_r , v_θ and their derivatives at $r = a$. [5]

(c) The relevant solutions to $\nabla^2 \omega = 0$ and $\nabla^2 p = 0$ take the form

$$\Psi = Cr \sin^2 \theta, \quad p = p_0 + \frac{\eta U a}{r^2} \cos \theta,$$

for constant p_0 . Show that the given form of Ψ can satisfy all boundary conditions. Show that the drag force on the bubble has magnitude

$$F = 4\pi\eta a U.$$

[7]

(d) Now consider a scenario where the air bubble mass M decreases due to dissolution into the liquid, with $dM/dt = -4\pi a D_a \rho_a$. The diffusivity D_a , air density ρ_a , water density ρ_w , air viscosity η_a and water viscosity η_w are all constant. If the air bubble initially has radius a_0 and rises due to buoyancy forces at a constant terminal velocity, determine how the bubble radius varies with height z through the water column. [4]

Ocean wave breaking carries air bubbles with $a_0 = 10^{-4}$ m to some depth d below the ocean surface. How large must d be if the bubbles are to entirely dissolve before they can escape back to the ocean surface? [2]

[You are given:

$$D_a \approx 10^{-10} \text{ m}^2 \text{ s}^{-1}, \quad \rho_a \approx 1 \text{ kg m}^{-3}, \quad \rho_w \approx 10^3 \text{ kg m}^{-3}, \quad \eta_w \approx 10^{-3} \text{ Pa s.}]$$